

Introduction

- In Mandarin, ***háishì*** forms alternative questions (AltQs):
[S_L ***háishì*** S_R]

(1) Ā Máo xǐhuān hē kāfēi (ne,) { *huòzhě/ ***háishì*** } xǐhuān hē chá (ne)?
A Mao like drink coffee NE or or like drink tea NE
‘Which one is true: A Mao likes drinking coffee or tea?’

- Previous analyses are **symmetric**:
háishì takes two arguments of the same type: two propositions, two questions, etc (Li & Thompson 1981, Huang 1991, Erlewine 2025, a.o.)

❖ Here I argue against this **symmetry**:

(2) $\llbracket \text{háishì} \rrbracket^w := \lambda p_{st}. \lambda Q_{stt}. \lambda r_{st}. [Q(r) \vee r = p]$
 $\equiv \lambda p. \lambda Q. [Q \cup \{p\}]$

- ***Háishì*** adds p to the Hamblin set of Q to form an augmented question Q' .

Observing asymmetry: [S_L ***háishì*** S_R] is sometimes [$Q?$ ***háishì*** p]

❑ Polar question + ***háishì*** + p

❖ [S_L ma] vs. * [S_R ma]

- S_L shows question properties,
- And S_R does not.

- Ma tracks the speech act of asking a PolQ.

(Li & Thompson 1981, Cheng 1991, Paul 2014, Pan 2019, a.o.)

(3) Matrix PolQ w/ ma

Fēi bù zhīdào Ā Máo hē le kāfēi ***ma***?
Fei NEG know A Mao drink ASP coffee QP
‘Does Fei not know that A Mao drank coffee?’

(4) * Wh -Q w/ ma

Ā Máo xǐhuān hē shénme ***ma***?
A Mao like drink what QP
#‘What does A Mao like drinking?’

- In AltQs, ma can only occur in the left disjunct S_L

(5) Ā Máo xǐhuān hē kāfēi ***ma***, ***háishì*** xǐhuān hē chá (ne)?
A Mao like drink coffee QP or like drink tea NE
‘Does A Mao like drinking coffee, or does he like drinking tea?’

(6) *Ā Máo xǐhuān hē kāfēi { ***ma***/ ne/ $\emptyset$ }, ***háishì*** xǐhuān hē chá ***ma***?
A Mao like drink coffee QP NE or like drink tea QP

❑ AltQ + ***háishì*** + p (i.e., recursion)

❖ [p (***háishì***) q ***háishì*** r]

- Assume it is [$[p$ (***háishì***) q] ***háishì*** r]
- ➔ [$AltQ$ ***háishì*** r] is available.

- More than two ps can be disjoined by ***háishì***.

(7) Ā Máo jīntiān (***háishì***) míngtiān ***háishì*** hòutiān lái (ne)?
A Mao today or tomorrow or in 2 days come NE
‘Will A Mao come today, tomorrow, or in two days?’

❑ Wh -question + ***háishì*** + p

❖ * [$\{ whQ, p \}$ ***háishì*** whQ]

- WhQ can only appear in S_L but not S_R

- When disjoining a whQ , it can only appear in S_L

(8) Nǐ dǎsuàn zài nǎr zū chē (ne), ***háishì*** dǎsuàn jiè yí-liàng (ne)? [$whQ + p$]
2sg plan LOC ***where*** rent car NE or plan borrow one-CL NE
‘Where do you plan to rent a car, or do you plan to borrow one?’ (after Ciardelli et al., 2018)

(9) Nǐ de diànhuà shì duōshǎo, { huòzhě /****háishì*** } nǐ de yóuxiāng shì shénme? [$whQ + whQ$]
2sg DE phone number be how many or or you DE email be what
‘(To recover the username, please tell me) what is your phone number, or what is your email?’

(10) Nǐ yǒu diànhuà ma, { huòzhě /****háishì*** } nǐ de yóuxiāng shì shénme? [$p + whQ$]
2sg have phone number QP or or you DE email be what
‘(To recover the username, please tell me) do you have a phone number, or what is your email?’

Hypothesis:

- ❖ The question in S_L provides the value of the second argument to ***háishì***.
(Assume ***LF***: [β [***háishì*** α]])

- The lexical entry in (2) spells out the semantic type asymmetry.

Challenge 1: Separate questions

- ❖ A question in S_L may be a separate sentence, anteceding a covert argument to ***háishì***:

Q1: [S_L] ? Q2 : [Δ ***Háishì*** S_R] ?

- Q2 is still an AltQ, since it resists polarity answers. E.g.,

(11) A: (5)? (12) A: (8)?
B: #Yes, he likes drinking tea. B: #Yes, we plan to borrow one.

- ❖ How to interpret Δ ? Is it a Q or a p ?

- One argument against Δ being propositional:

- When S_L is a PolQ: Δ = the positive answer p of Q1.
- Otherwise (7)/(8): Δ = the existential presupposition of Q1.

➤ **Unattractive**

Two distinct ways needed for anaphoric interpretation.

- If Δ is a Q , its value is just Q1, and (2) can be maintained.

Challenge 2: Example (1)

- ❖ Proposal in (2) predicts S_L in (1) to be a question. Evidence?

- Declarative strings can be used to ask PolQs, if with rising prosody.

(13) Ā Máo xǐhuān hē kāfēi
A Mao like drink coffee
(I) ‘Does A Mao like drinking coffee?’
(D) ‘A Mao likes drinking coffee.’

- ❖ If S_L were a proposition, how to handle recursion like (7)?

- If (7) involves separate questions, then propositional anaphora is **disfavored**.

- If (7) results from a recursive composition, then we need the output of ***háishì*** to be a proposition.

- Otherwise, if ***háishì*** outputs a Q , then the Q would not feed into composition with another ***háishì***.

Discussion

Two analyses where ***háishì*** takes two propositions

- ❖ Li & Thompson (1981): ***Háishì*** = C_Q + or

(14) $\llbracket \text{háishì} \rrbracket = \lambda p_{st}. \lambda q_{st}. \lambda r_{st}. r = p \vee r = q.$

- The output would not account for the recursion fact.

- ❖ Huang (1991): ***Háishì*** = or

(15) $\llbracket \text{háishì} \rrbracket = \lambda p_t. \lambda q_t. p \vee q.$

- ***Háishì*** is a logical connective.
- Assuming Karttunen (1977), the interrogative reading results from scoping ***háishì*** over C_Q .

- It might overgenerate an unattested LF (e.g., (9))

(16) *LF: [Op_1 [$what_3$ [$[C_Q t_1]$ your-number-is t_3] ***háishì*** _{$\langle t, \langle t, t \rangle \rangle$} [$what_4$ [$[C_Q t_1]$ your-email-is t_4]]]]

Declarative use of ***háishì***

- ❖ My proposal assumes lexical ambiguity (cf. Erlewine 2025)

(17) Ā Máo hǎoxiàng hē le kāfēi ***háishì*** chá.
A Mao seem drink ASP coffee or tea
‘It seems that A Mao drank coffee or tea.’